1.1 AI Version Used

ChatGPT 6

1.2 Date Used

2 November 2026

1.3 How AI Helped with the Assignment

I asked ChatGPT to explain the GPS assignment requirements and how to divide the implementation into small steps. It helped outline the GUI, ten tracker streams, filtering, altitude conversion, five-minute distances, testing, and submission evidence. I compared the plan with the assignment instructions and worked on the Java project in stages.

2.1 AI Version Used

ChatGPT 6

2.2 Date Used

2 November 2026

2.3 How AI Helped with the Assignment

I asked how to compile and run the Java GUI with the Sodium JAR in Windows PowerShell. It suggested classpath commands using the supplied sodium-1.2.0.jar and the provided GPS classes. I used the commands locally and checked the GUI could start.

3.1 AI Version Used

ChatGPT 6

3.2 Date Used

2 November 2026

3.3 How AI Helped with the Assignment

I asked ChatGPT how to organise the GPS GUI so the ten trackers, distance values, filters and latest event would be easy to understand. It explained how Swing panels and layouts could group related information. AI also provided example code during development. I checked the running layout and made sure the relevant information was visible.

4.1 AI Version Used

ChatGPT 6

4.2 Date Used

2 November 2026

4.3 How AI Helped with the Assignment

I asked how Sodium map could be used to show just the tracker number, latitude and longitude. ChatGPT explained how a stream transformation can remove the altitude from a simplified display and showed implementation examples. I checked the ten tracker rows in the GUI.

5.1 AI Version Used

ChatGPT 6

5.2 Date Used

3 November 2026

5.3 How AI Helped with the Assignment

I asked how to combine the ten separate GPS streams in a reactive design. ChatGPT explained orElse and how to use it to build a combined stream, with example implementation. I checked the combined-stream logic in GpsGUI.java and noted that simultaneous events may need further testing.

6.1 AI Version Used

ChatGPT 6

6.2 Date Used

4 November 2026

6.3 How AI Helped with the Assignment

I asked how to design a filter that changes while GPS events continue arriving. ChatGPT explained CellSink, snapshot and filter, and helped with example implementation. I tested the settings -90/-180 and 39.9/116.3 in the GUI to check the accepted-event display.

7.1 AI Version Used

ChatGPT 6

7.2 Date Used

4 November 2026

7.3 How AI Helped with the Assignment

I asked how to check invalid user input before updating the filter. ChatGPT explained numeric parsing, coordinate ranges and error handling, and provided example code. I tested an invalid latitude of abc and saw the Invalid Input message.

8.1 AI Version Used

ChatGPT 6

8.2 Date Used

3 November 2026

8.3 How AI Helped with the Assignment

I asked how to display the latest accepted GPS event in a simple four-value format. ChatGPT explained how the tracker name, latitude, longitude and original altitude could be joined for display and helped with implementation. I checked the label while GPS data was arriving.

9.1 AI Version Used

ChatGPT 6

9.2 Date Used

3 November 2026

9.3 How AI Helped with the Assignment

I asked how to design a three-second timeout for the latest GPS event. ChatGPT explained using a non-repeating Swing Timer and restarting it after each new accepted event, and provided implementation help. I inspected the timer logic and later ran a separate component demonstration.

10.1 AI Version Used

ChatGPT 6

10.2 Date Used

4 November 2026

10.3 How AI Helped with the Assignment

I asked why GPS altitude in feet must be converted and how Sodium map could help. ChatGPT explained the fixed conversion factor 0.3048 and showed code examples for metric GPS events. I checked the displayed metres and the conversion test.

11.1 AI Version Used

ChatGPT 6

11.2 Date Used

5 November 2026

11.3 How AI Helped with the Assignment

I asked how to calculate movement distance from changes in GPS latitude, longitude and altitude. ChatGPT explained how to approximate horizontal movement in metres and combine it with vertical distance. It also provided calculation examples. I checked the distance tests for several movement directions.

12.1 AI Version Used

ChatGPT 6

12.2 Date Used

6 November 2026

12.3 How AI Helped with the Assignment

I asked how to design a moving five-minute window for each tracker. ChatGPT explained timestamps, per-tracker position lists, and how to remove old positions, with example code. I checked the exact five-minute boundary and tracker-independence tests.

13.1 AI Version Used

ChatGPT 6

13.2 Date Used

7 November 2026

13.3 How AI Helped with the Assignment

I asked how to represent distance history using Sodium rather than only ordinary mutable variables. ChatGPT explained StreamSink commands and collect for updating state, and helped with code examples for ADD, TICK and RESET. I reviewed this section and used automated state tests.

14.1 AI Version Used

ChatGPT 6

14.2 Date Used

7 November 2026

14.3 How AI Helped with the Assignment

I asked how to avoid mixing GPS events from before and after a filter change. ChatGPT explained the idea of versioning and resetting histories, with examples. I checked the GUI reset behaviour and tests that reject events from previous filter versions.

15.1 AI Version Used

ChatGPT 6

15.2 Date Used

7 November 2026

15.3 How AI Helped with the Assignment

I asked what tests would be useful for distance, altitude, filtering, resetting and rounding. ChatGPT helped design and write a test class with 12 cases. I ran it locally and checked the output: 12 passed and 0 failed.

16.1 AI Version Used

ChatGPT 6

16.2 Date Used

7 November 2026

16.3 How AI Helped with the Assignment

I asked how to test Sodium operations with zero, one and two simulated GPS events. ChatGPT explained the testing approach and helped write stream tests for map, merging, snapshot and filtering. I ran the test locally and checked that 10 cases passed.

17.1 AI Version Used

ChatGPT 6

17.2 Date Used

8 November 2026

17.3 How AI Helped with the Assignment

I asked what boundary conditions could cause mistakes in the five-minute distance calculation. ChatGPT suggested checks for exact cutoff time, expired positions, separate trackers, resets and old filter versions, and helped implement the tests. I ran them and checked that 6 passed.

18.1 AI Version Used

ChatGPT 6

18.2 Date Used

8 November 2026

18.3 How AI Helped with the Assignment

I asked how to demonstrate the three-second clearing behaviour without pressing Set Filter. ChatGPT explained how a separate test window could use a Swing Timer and supplied a complete example test program. I compiled and ran it and observed the event disappear after the timer callback. This was a separate component test, not a full-GUI test.

19.1 AI Version Used

ChatGPT 6

19.2 Date Used

8 November 2026

19.3 How AI Helped with the Assignment

I asked ChatGPT how to organise my testing evidence and explain the main FRP operations in a video. It helped draft the testing report and simple English speaking notes. I checked the report against my screenshots, changed its formatting to plain black text, and used the line numbers in GpsGUI.java to prepare my explanation.